

Dietary Fat and Cholesterol and Risk of Cardiovascular Disease in Older Adults: the Health ABC Study

BACKGROUND AND AIMS Although dietary fats and cholesterol have previously been associated with risk of cardiovascular disease (CVD) in middle aged populations, less is known among older adults. The purpose of this study was to determine the association between dietary fats, cholesterol, and eggs and CVD risk among community-dwelling adults aged 70–79 in the Health, Aging and Body Composition Study.

METHODS AND RESULTS Diet was assessed using an interviewer-administered 108-item food frequency questionnaire (n=1,941). CVD events were defined as a confirmed myocardial infarction, coronary death, or stroke. Relative rates of CVD over 9 years of follow-up were estimated using Cox proportional hazards models. During follow-up, there were 203 incident cases of CVD. There were no significant associations between dietary fats and CVD risk. Dietary cholesterol (HR (95% CI): 1.47 (0.93, 2.32) for the upper vs. lower tertile; P for trend, 0.10) and egg consumption (HR (95% CI): 1.68 (1.12, 2.51) for 3+/week vs. <1/week); P for trend, 0.01) were associated with increased CVD risk. However, in subgroup analyses, dietary cholesterol and egg consumption were associated with increased CVD risk only among older adults with type 2 diabetes (HR (95% CI): 3.66 (1.09, 12.29) and 5.02 (1.63, 15.52), respectively, for the upper vs. lower tertile/group).

CONCLUSIONS Dietary cholesterol and egg consumption were associated with increased CVD risk among older, community-dwelling adults with type 2 diabetes. Further research on the biological mechanism(s) for the increased CVD risk with higher dietary cholesterol and frequent egg consumption among older adults with diabetes is warranted.